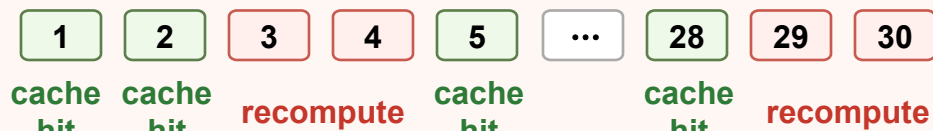


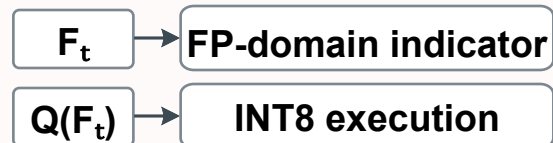
Challenges

1 Decision-Domain Mismatch in Cache+Quant

Illustrative Denoising timeline (30 steps) ■ cache hit ■ recompute



FP indicator $\neq$ INT8 execution relevance



2 Matrix-Axis Mismatch in Factorized Attention

Spatial: 38 (19x2) contexts $\times$ 1024-token rows



Temporal: 2048 (1024x2) contexts $\times$ 19-token rows



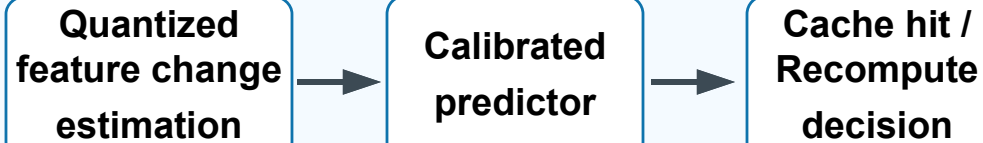
54 $\times$ more contexts

QK^T : $[1, d] \times [d, L] \rightarrow$ irregularity on output axis

PV: $[1, L] \times [L, d] \rightarrow$ irregularity on reduction axis

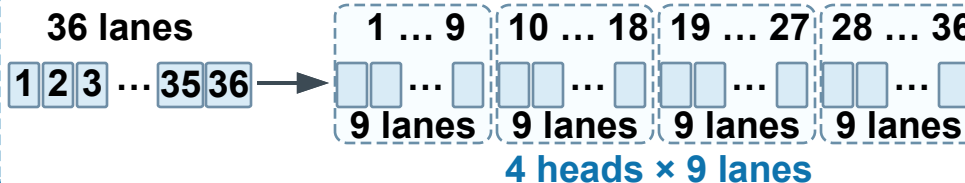
Proposed Co-Design

1 QAC: Quantization-Aligned Caching

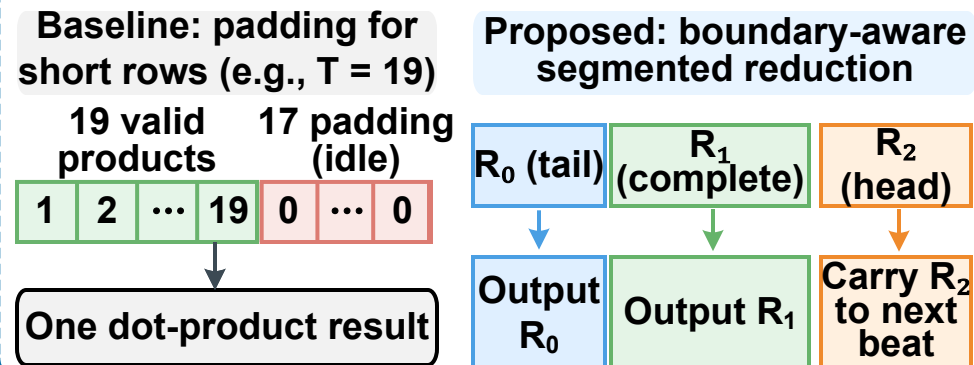


2 DARE: Dimension-Adaptive Reconfigurable Engine

① QK output-axis adaptation

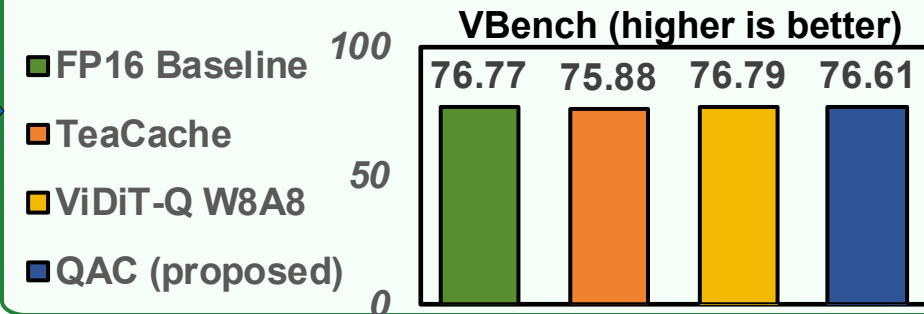


② PV reduction-axis adaptation



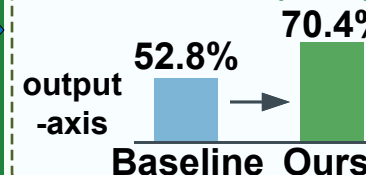
Benefits / Results

1 Algorithm Quality

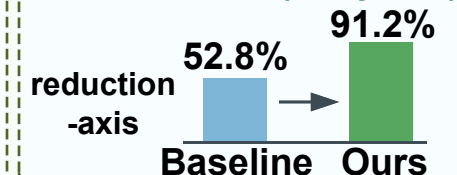


2 Hardware Efficiency

QK utilization (Temporal)



PV utilization (temporal)



3 System-Level Gains

Throughput : 10.96 $\times$ vs A100; 1.06 $\times$ vs ViDA

Energy Efficiency : 374.7 $\times$ vs A100

Area Efficiency : 24.07 $\times$ vs ViDA